The Method of Moderation

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Abstract

In a risky world, a pessimist assumes the worst will happen, while someone who ignores risk altogether is an optimist. Consumption is mathematically simple for both, since each behaves as if the world were riskless. A realist, who responds optimally to risk, faces a much harder problem but (under standard conditions) spends somewhere between the pessimist and the optimist. We use this fact to redefine the space in which the realist searches for optimal consumption rules. The resulting solution accurately represents the consumption rule over the entire range of feasible wealth with remarkably few computations.

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JEL classification: D14; C61; G11

1. Introduction

Solving a consumption-saving problem using numerical methods requires the modeler to choose how to represent a policy function. In the stochastic case, where analytical solutions are generally not available, a common approach is to use low-order polynomial splines that exactly match the function at a finite set of gridpoints, and then to assume that interpolated or extrapolated versions of that spline represent the function well at the continuous infinity of unmatched points. Carroll (2006) developed the endogenous gridpoints method (EGM), which has become a standard tool for computing consumption at gridpoints determined endogenously using the Euler equation.

Unfortunately, this endogenous gridpoints solution is not very well-behaved outside the original range of gridpoints, though other common solution methods are no better outside their own predefined ranges. Figure 1 demonstrates

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the point. The figure shows the approximated precautionary component of saving, the amount by which the realist consumes less than an optimist with the same expected income path. Theory proves that precautionary saving is always positive, yet the linearly extrapolated numerical approximation eventually predicts negative precautionary saving. Under uncertainty, however, the consumption-saving rule must be evaluated outside *any* prespecified grid, because large positive shocks push a sufficiently wealthy individual's next-period assets beyond the grid boundaries.

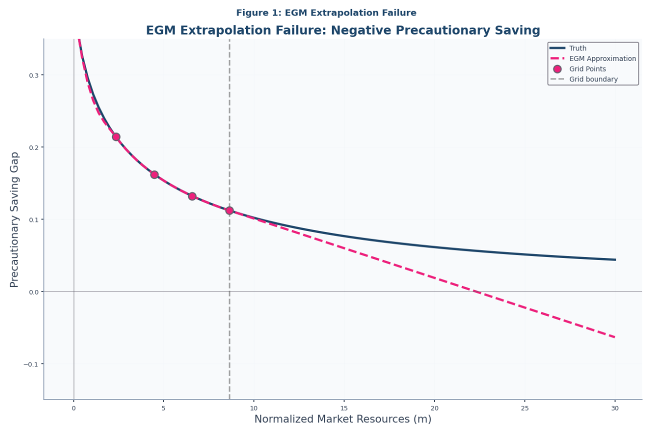

Figure 1: For Large Enough Resources m, Predicted Precautionary Saving is Negative (Oops!)

This error cannot be fixed by extending the upper gridpoint. While extrapolation techniques can prevent this from being fatal, the problem is often dealt with using inelegant methods whose implications for accuracy are difficult to gauge.

This paper argues that, in the standard consumption problem, a better approach is to rely upon the fact that without uncertainty, the optimal consumption function has a simple analytical solution. The key insight is that, under standard assumptions, the consumer who faces an uninsurable labor income risk will consume less than a consumer with the same path for expected income but who does not perceive any uncertainty. Following Leland (1968), Sandmo (1970), and Kimball (1990), the 'realist' consumer, who *does* perceive the risks, will engage in 'precautionary saving,' so the perfect foresight riskless solution provides an upper bound to the solution that will actually be optimal. A lower bound is provided by the behavior of a consumer who has the subjective belief that the future level of income will be the worst that it can possibly be. This consumer, too, behaves according to the convenient analytical perfect foresight solution, but his certainty is that of a pessimist perfectly confident in his pessimism.

We build on bounds for the consumption function and limiting MPCs established by Stachurski and Toda (2019), Ma et al. (2020), Carroll (2009), and Ma and Toda (2021) in buffer-stock theory. Using results from Carroll and

Shanker (2024), we show how to use these bounds to constrain the shape and characteristics of the solution to the 'realist' problem characterized by Carroll (1997). Imposition of these constraints clarifies and speeds the solution of the realist's problem. For comparison, we use the endogenous gridpoints method (Carroll, 2006) as our benchmark, which computes consumption at gridpoints determined endogenously using the Euler equation.

After presenting the method in the baseline case and documenting its accuracy, we collect its refinements and extensions, a tighter theoretical bound, the value function, and an extension to rate-of-return risk, in the appendix.

2. The Realist's Problem

Consider a consumer who correctly perceives all risks. The consumer has CRRA utility over consumption c with risk aversion parameter $\rho > 0$:

$$\mathbf{u}(c) = \begin{cases} \frac{c^{1-\rho}}{1-\rho} & \text{if } \rho \neq 1 \\ \log c & \text{if } \rho = 1. \end{cases} \tag{1}$$

This utility function satisfies prudence ($\mathbf{u}''' > 0$), which induces precautionary saving. The consumer maximizes expected lifetime utility, discounted by the time preference factor β:

$$\max_{c_t, a_t} \mathbb{E}_t \left[\sum_{n=0}^{T-t} \beta^n \mathbf{u}(c_{t+n}) \right] \tag{2}$$

subject to $c_t + a_t = m_t$, where m denotes market resources and a denotes assets. We focus on resources of the form $m_{t+1} = a_t \mathrm{R}_{t+1} + y_{t+1}$, where R_{t+1} denotes the interest rate, and y_{t+1} labor income. Initially we take R_{t+1} to be deterministic, and relax this later.

While our method can be adapted to a range of stochastic labor income processes, to fix ideas we suppose income evolves via the Friedman-Muth process (Friedman (1957) distinguished permanent from transitory income; Muth (1960) provided the stochastic framework). That is, $y_{t+1} = p_{t+1}\xi_{t+1}$ where p denotes permanent labor income and ξ_{t+1} a transitory component. Permanent income growth is given by $p_{t+1} = p_t \mathcal{G}_{t+1}$, where the gross growth factor $\mathcal{G}_{t+1} = G_{t+1}\psi_{t+1}$. Here G_{t+1} is the deterministic permanent income growth factor, while ψ_{t+1} are permanent shocks with mean unity and bounded support $[\underline{\psi}, \bar{\psi}]$ where $0 < \underline{\psi} \leq 1 \leq \bar{\psi} < \infty$. Transitory shocks ξ_{t+1} take value 0 with probability $\wp > 0$ (unemployment) or $\theta_{t+1}/(1-\wp)$ otherwise, with $\mathbb{E}_t[\theta_{t+1}] = 1$.

This problem can be rewritten (see Carroll (2020) for a proof) in a more convenient form in which choice and state variables are normalized by the level of permanent income, e.g., replacing m_t with m_t/p_t. When that is done, the Bellman equation for the value function $\mathbf{v}$ of the transformed problem is

$$\mathbf{v}_t(m_t) = \max_{c_t, a_t} \left(\mathbf{u}(c_t) + \beta \mathbb{E}_t[\mathcal{G}_{t+1}^{1-\rho} \mathbf{v}_{t+1}(m_{t+1})] \right) \tag{3}$$

subject to the normalized transition $m_{t+1} = (\mathrm{R}/\mathcal{G}_{t+1})a_t + \xi_{t+1}$, with Euler equation $\mathbf{u}'(c_t) = \beta\mathrm{R}\mathbb{E}_t[\mathcal{G}_{t+1}^{-\rho}\mathbf{u}'(c_{t+1})]$.

Carroll and Shanker (2024) gives conditions for a finite solution of the problem with a Friedman-Muth process. Consider the case of time-invariant G_t, ψ_t, and R_t, and define the absolute patience factor $\Phi \equiv (\beta\mathrm{R})^{1/\rho}$. Then a finite solution requires: (i) finite-value-of-autarky condition (FVAC) $0 < \beta G^{1-\rho}\mathbb{E}[\psi^{1-\rho}] < 1$, (ii) absolute-impatience condition (AIC) $\Phi < 1$, (iii) return-impatience condition (RIC) $\Phi/\mathrm{R} < 1$, (iv) growth-impatience condition (GIC) $\Phi/G < 1$, and (v) finite-human-wealth condition (FHWC) $G/\mathrm{R} < 1$. These patience conditions ensure the consumption bounds and limiting MPCs used in our method.

For expositional simplicity, in the numerical results that follow, we assume $\rho \neq 1$ and set $G = 1$, $\psi = 1$ (no permanent income growth or shocks). The figures display the next-to-last period of a finite horizon problem, where the last-period analytical solution provides exact boundary conditions. The method extends naturally to the infinite-horizon case and to general parameter configurations.

3. The Method of Moderation

3.1. The Optimist, the Pessimist, and the Realist

As a preliminary to our solution, define $\bar{h}$[1] as end-of-period human wealth (the present discounted value of future labor income) for a perfect foresight version of the problem of a 'risk optimist:' a consumer who believes with perfect confidence that the shocks will always take their expected value of 1, $\xi_{t+n} = \mathbb{E}[\xi] = 1 \; \forall \; n > 0$. The solution to a perfect foresight problem of this kind takes the form

$$\bar{\mathbf{c}}(m) = (m + \bar{h})\underline{\boldsymbol{\kappa}} \tag{4}$$

for a constant minimal marginal propensity to consume $\underline{\boldsymbol{\kappa}}$.[2] We similarly define $\underline{h}$[3] as 'minimal human wealth,' the present discounted value of labor income if the shocks were to take on their worst value in every future period $\xi_{t+n} = \underline{\xi} \; \forall \; n > 0$. We refer to a consumer who expects to encounter this sequence of shocks as a 'pessimist'. Their consumption decision rule is given by

$$\underline{\mathbf{c}}(m) = (m + \underline{h})\underline{\boldsymbol{\kappa}}. \tag{5}$$

We will call a 'realist' the consumer who correctly perceives the true probabilities of the future risks and optimizes accordingly.

[1] Setting $\xi_{t+n} = 1$ (the optimist's assumption), human wealth in infinite-horizon is $\bar{h} = G/(\mathrm{R} - G)$ if $\mathrm{R} > G$. When $G = 1$, $\bar{h} = 1/(\mathrm{R} - 1)$.

[2] The MPC of the perfect foresight consumer: infinite-horizon $\underline{\boldsymbol{\kappa}} = 1 - \Phi/\mathrm{R}$.

[3] Setting $\xi_{t+n} = \underline{\xi} \; \forall \; n > 0$, minimal human wealth is $\underline{h} = \underline{\xi}G/(\mathrm{R} - G)$ if $\mathrm{R} > G$. When $\underline{\xi} = 0$, $\underline{h} = 0$.

A first useful point is that, for the realist, a lower bound for the level of market resources is the natural borrowing constraint $\underline{m} = -\underline{h}$ derived by Aiyagari (1994) and Huggett (1993), because if m equalled this value then there would be a positive finite chance (however small) of receiving $\xi_{t+n} = \underline{\xi}$ in every future period, which would require the consumer to set c to zero in order to guarantee that the intertemporal budget constraint holds. Since consumption of zero yields infinite marginal utility, Zeldes (1989) and Deaton (1991) show that the solution to the realist consumer's problem is not well defined for values of $m \leq \underline{m}$, and the limiting value of the realist's c is zero as $m \downarrow \underline{m}$ (established in Carroll and Shanker (2024)).

It is convenient to define 'excess' market resources as the amount by which actual resources exceed the lower bound, and 'excess' human wealth as the amount by which mean expected human wealth exceeds guaranteed minimum human wealth:

$$
\begin{aligned}
\Delta m &= m + \overbrace{\underline{h}}^{=-\underline{m}} \\
\Delta h &= \bar{h} - \underline{h}.
\end{aligned}
\tag{6}
$$

We now rewrite the optimal consumption rules for the two perfect foresight problems in terms of excess resources and human wealth. The 'pessimist' perceives human wealth to be equal to its minimum feasible value $\underline{h}$ with certainty, and so consumes a constant fraction of current excess resources

$$\underline{\mathbf{c}}(m) = \Delta m \underline{\boldsymbol{\kappa}}. \tag{7}$$

The 'optimist,' on the other hand, pretends that there is no uncertainty about future income, and therefore consumes the same fraction out of current excess resources *and* excess human wealth

$$\bar{\mathbf{c}}(m) = (\Delta m + \Delta h)\underline{\boldsymbol{\kappa}}. \tag{8}$$

3.2. The Moderation Ratio

The pessimist expects the worst possible income in every future period with certainty, and must finance all future consumption through saving; this generates the most aggressive saving behavior. A realist, by contrast, can reoptimize as uncertainty resolves each period, so need not prepare for the worst with certainty. At the same time, the adverse outcome remains possible, so even a wealthy realist maintains some precautionary saving: a sufficiently well-off individual is nearly (but never completely) self-insured, and thus mostly smooths consumption like the optimist. The realist therefore consumes strictly more than the pessimist but strictly less than the optimist at every wealth level, as shown in Figure 2.

The proof is more difficult than might be imagined, but the necessary work is done in Carroll and Shanker (2024) so we will take the proposition as a fact:

$$\underline{\mathbf{c}}(\underline{m} + \Delta m) < \hat{\mathbf{c}}(\underline{m} + \Delta m) < \bar{\mathbf{c}}(\underline{m} + \Delta m) \tag{9}$$

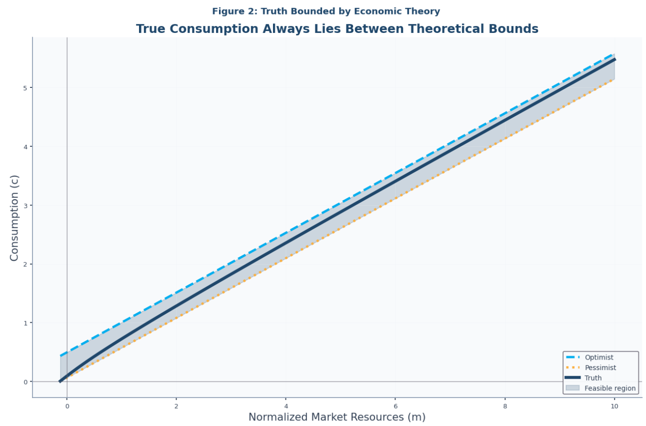

Figure 2: Moderation Illustrated: $\underline{\mathbf{c}} < \hat{\mathbf{c}} < \bar{\mathbf{c}}$

Subtracting $\underline{\mathbf{c}}(\underline{m}+\Delta m)$ in each of these inequalities and using equations (7) and (8) gives

$$\begin{aligned}
0 < \quad & \hat{\mathbf{c}}(\underline{m}+\Delta m) - \underline{\mathbf{c}}(\underline{m}+\Delta m) & < \Delta h \underline{\boldsymbol{\kappa}} \\
0 < \quad & \underbrace{\left(\frac{\hat{\mathbf{c}}(\underline{m}+\Delta m) - \underline{\mathbf{c}}(\underline{m}+\Delta m)}{\Delta h \underline{\boldsymbol{\kappa}}}\right)}_{\equiv \boldsymbol{\omega}} & < 1
\end{aligned} \tag{10}$$

where the fraction in the middle of the last inequality is the moderation ratio measuring how close the realist's consumption is to the optimist's behavior (the numerator is the gap between the realist and pessimist) relative to the maximum possible gap between optimist and pessimist. When $\boldsymbol{\omega} = 0$, the realist behaves like the pessimist (maximum precautionary saving); when $\boldsymbol{\omega} = 1$, the realist behaves like the optimist (no precautionary saving). Carroll and Kimball (1996) and Carroll and Shanker (2024) establish that under bounded shocks, $\boldsymbol{\omega} \in (0, 1)$ strictly for all $m > \underline{m}$. Defining $\boldsymbol{\mu} = \log \Delta m$ (which can range from $-\infty$ to ∞), the object in the middle of the last inequality is

$$\boldsymbol{\omega}(\boldsymbol{\mu}) \equiv \left(\frac{\hat{\mathbf{c}}(\underline{m}+e^{\boldsymbol{\mu}}) - \underline{\mathbf{c}}(\underline{m}+e^{\boldsymbol{\mu}})}{\Delta h \underline{\boldsymbol{\kappa}}}\right), \tag{11}$$

and we now define

$$\begin{aligned}
\boldsymbol{\chi}(\boldsymbol{\mu}) &= \log\left(\frac{\boldsymbol{\omega}(\boldsymbol{\mu})}{1-\boldsymbol{\omega}(\boldsymbol{\mu})}\right) \\
&= \log(\boldsymbol{\omega}(\boldsymbol{\mu})) - \log(1-\boldsymbol{\omega}(\boldsymbol{\mu}))
\end{aligned} \tag{12}$$

which has the virtue that it is *asymptotically linear* in the limit as $\boldsymbol{\mu}$ approaches $+\infty$.[4] Since $\boldsymbol{\omega} \in (0, 1)$, the ratio $\boldsymbol{\omega}/(1-\boldsymbol{\omega})$ is the odds ratio, and $\boldsymbol{\chi}$ is

[4]Under the GIC, $\boldsymbol{\chi}(\boldsymbol{\mu})$ is asymptotically linear with slope $\alpha \geq 0$ as $\boldsymbol{\mu} \to +\infty$. We ex-

the log odds ratio, the same transformation that underpins logit regression in econometrics. The logit maps $\boldsymbol{\omega} \in (0,1)$ to $\boldsymbol{\chi} \in (-\infty, \infty)$ with inverse sigmoid $\boldsymbol{\omega} = 1/(1+\exp(-\boldsymbol{\chi}))$; log maps $(m - \underline{m}) \in (0, \infty)$ to $\boldsymbol{\mu} \in (-\infty, \infty)$ with inverse $\Delta m = \exp(\boldsymbol{\mu})$.

Given $\boldsymbol{\chi}$, the consumption function can be recovered from

$$\hat{\mathbf{c}} = \underline{\mathbf{c}} + \overbrace{\frac{1}{1+\exp(-\boldsymbol{\chi})}}^{=\boldsymbol{\omega}} \Delta h \underline{\boldsymbol{\kappa}}. \tag{13}$$

Thus, the method of moderation is to calculate $\boldsymbol{\chi}$ at the points $\boldsymbol{\mu}$ corresponding to the log of the Δm points defined above, and then using these to construct an interpolating approximation $\grave{\boldsymbol{\chi}}$ from which we indirectly obtain our approximated consumption rule $\grave{\mathbf{c}}$ (an approximation to the true $\hat{\mathbf{c}}$) by substituting $\grave{\boldsymbol{\chi}}$ for $\boldsymbol{\chi}$ in equation (13).

Because this method relies upon the fact that the problem is easy to solve if the decision maker has unreasonable views (either in the optimistic or the pessimistic direction), and because the correct solution is always between these immoderate extremes, we call our solution procedure the 'method of moderation.'

Results are shown in Figure 3; a reader with very good eyesight might be able to detect the barest hint of a discrepancy between the Truth and the Approximation at the far right-hand edge of the figure, a stark contrast with the calamitous divergence evident in Figure 1.

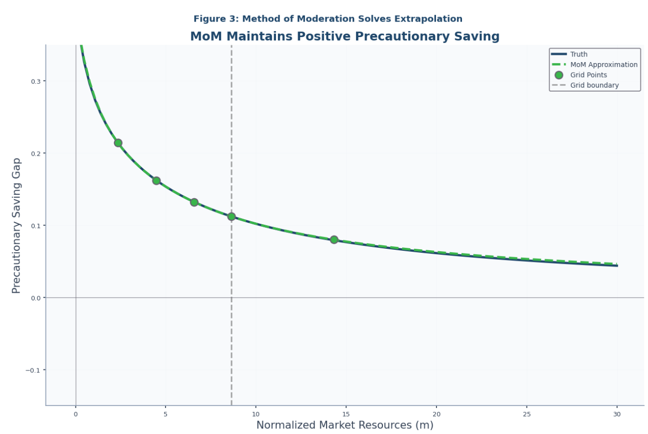

Figure 3: Extrapolated $\grave{\mathbf{c}}$ Constructed Using the Method of Moderation

trapolate $\boldsymbol{\chi}$ linearly using the boundary slope, preserving $\boldsymbol{\omega} \in (0,1)$ and hence $\underline{\mathbf{c}} < \grave{\mathbf{c}} < \bar{\mathbf{c}}$ throughout the extrapolation domain.

Table 1: Maximum absolute approximation errors by interval. Orders of magnitude in parentheses.

Method	$[m_0, m_1]$	$[m_1, m_2]$	$[m_2, m_3]$	$[m_3, m_4]$	$[m_4, \overline{m}]$
EGM	8.6(-3)	1.8(-4)	2.5(-5)	7.3(-6)	1.1(-1)
MoM	2.9(-3)	4.3(-6)	6.6(-7)	1.3(-7)	2.4(-3)

4. Numerical Accuracy

Table 1 reports the method's accuracy between each pair of gridpoints m_j, m_{j+1} and for extrapolation out to $\overline{m} = 30$, as the maximum absolute error on a dense subgrid of each interval. Except in the first interval, where it is about three times more accurate, the method of moderation is more than an order of magnitude more accurate than the basic endogenous gridpoints method, complementing related work on solution precision (Chipeniuk, 2020).

The construction extends further in the Appendix. A tighter upper bound holds near the natural borrowing constraint for low-wealth consumers ((**??**)). The same idea approximates the value function: with the inverse value transformation $\bar{\mathbf{\Lambda}} = ((1-\rho)\bar{\mathbf{v}})^{1/(1-\rho)}$, we form a value moderation ratio $\hat{\mathbf{\Omega}}$, interpolate its logit, and invert to recover $\hat{\mathbf{v}} = \mathbf{u}(\hat{\mathbf{\Lambda}})$ ((**??**)). It also accommodates rate-of-return risk: with i.i.d. returns and certain income the optimal rule is linear (Merton-Samuelson), so the optimist's and pessimist's bounds extend and bracket the realist (Stochastic Rate of Return).

5. Conclusion

The method is not universally applicable: it requires known upper and lower bounds on the 'true' solution. But many problems have obvious bounds, and there (as in our consumption example) it can substantially improve the accuracy and stability of solutions. The method is efficient because the transformed moderation ratio is better-behaved than consumption, so a given grid yields substantially more accurate approximations. The gains are largest in the extrapolation region, where standard methods fail most severely.

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